Study on the velocity compensated method applied for the CSL Mode Suppression of CPW Bend

Hui Zhang[*1]*, Zengrui Li*[2]*, and Junhong Wang*[3,4]

[1]Information Engineering School, Communication University of China, Beijing 100024, China, zhangh@cuc.edu.cn

[2]Information Engineering School, Communication University of China, Beijing 100024, China, zrli@cuc.edu.cn

[3]Key Laboratory of All Optical Network & Advanced Telecommunication Network of EMC, Beijing JiaoTong University, Beijing 100044, China, wangjunh@bjtu.edu.cn

[4]Institute of Lightwave Technology, Beijing JiaoTong University, Beijing 100044, China, wangjunh@bjtu.edu.cn

Abstract

In the CPW bend, two slots with different length will excite radiation due to the coupled slot line (CSL) mode. In order to suppress this radiation and keep signal in the two slots in phase, the velocity compensation method is studied in this paper, in which the velocity of signal propagating in the shorter slot is slowed down or speeded up in the longer slot. the CPW bends with three kinds of compensation techniques based on this method are studied and compared. The FDTD method and commercial software are employed to analyze these CPW bends, the results are compared with those of air-bridged CPW bends. Research results show that the CSL mode can be suppressed efficiently by the velocity compensated method, and the transmission performance is also improved significantly.

1. Introduction

Coplanar waveguide (CPW) has been extensively used in monolithic microwave integrated circuits (MMIC's) due to its advantages in low dispersion, low radiation, ease of shunt-series connections and integration. Nevertheless, there are two modes in CPW, which are the CPW mode and CSL mode, also called even and odd modes, respectively. The CPW mode is the dominate mode. The CSL mode will cause radiation, and must therefore be eliminated. CSL mode always occurs in asymmetric CPW due to the distance gap between two slots. Many published works about CSL mode suppression of the asymmetric CPW include using air-bridges [1-3], chamfering the bend corner [4], and using top and/or bottom ground plane shields [5, 6]. As we know, the air-bridges are widely applied in the production of CPW. However, air-bridge technique needs additional fabrication procedure to bridge the two ground planes of CPW, especially this fabrication procedure will be more complicated where a large number of air-bridges are needed in MMIC. In this paper, a new method is introduced in which the phase difference of signals due to distance gap between the two slots is compensated by the speed gap of the signals propagating in those two slots. Two kinds of velocity compensated structure using grating and punching technique is studied in our previous work [7,8], and the advantage of these structures is proved both by simulation and experiment results. The third velocity compensated structure employing capacitance loading technique is taken as examples in this paper. Besides, the simulation results demonstrate that the CSL mode can be suppressed and the transmission property of CPW bends can be improved after using the proposed method.

2. Theory and Method

In conventional CPW bend, as shown in Fig. 1(a), the pulse excited from points A and A′ will propagate in the two slots with different length, so that the signal will arrive at points B and B′ at the different time. After the bend, there will be phase difference between signals in the two slots, the CSL mode will subsequently occur in the two slots. Assume that V_1 and V_2 are the voltages between the central conductor and the two ground planes, and positive direction of the voltage is defined from the central conductor to the two ground planes, respectively. Then, CPW mode voltage V_e and CSL mode voltage V_o can be expressed as

$$V_e = \frac{V_1 + V_2}{2}, \quad V_o = \frac{V_1 - V_2}{2} \tag{1}$$

From equation (1) we can see, the voltage of CSL mode V_o equal to zero in the two symmetrical slots before CPW bend, but the CSL mode voltage will be very large in the two vertical slots after the bend. The electric fields excited by V_o are in phase in two slots, so the CSL mode radiation in those two slots will superimpose each other. As we know, in order to suppress the CSL mode, V_o must equal to zero. Assume that V_i is the input voltage, S is the length of inner slot, ΔS is the path difference between the inner and outer slots, and the wave number of the two slots is k_1. For the case that the compensation method is employed in the inner slot, the wave number of this slot is assumed to be k_2, V_o become:

978-1-4673-5225-3/14/$31.00 ©2014 IEEE

$$V_o = \frac{1 - e^{-jk_1(S+\Delta S)+jk_2S}}{2} V_i e^{-jk_2S} \tag{2}$$

If let $V_o = 0$, we have

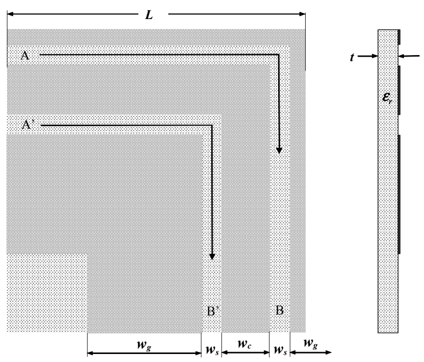
$$k_1(S + \Delta S) = k_2 S \Rightarrow S = \frac{\Delta S}{k_2 / k_1 - 1} \tag{3}$$

If equation (4) is true, must be satisfied: $k_2 > k_1$. Therefore, CSL mode is suppressed through enlarging the value of wave number k in the inner slot. Because k is inverse ratio to the signal velocity v, v in the inner slot needs to be slowed down to suppress the CSL mode. Otherwise, if the outer slot is compensated and the wave number of this slot is assumed to be k_2', in the same manner, we have

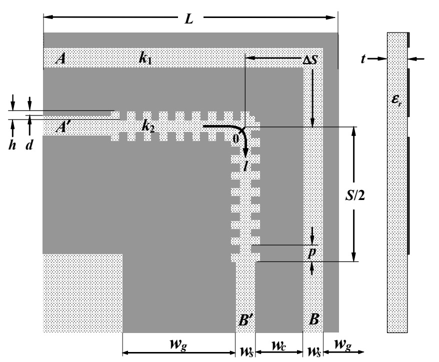
$$S = \frac{\Delta S}{k_1 / k_2^{'} - 1} \rightarrow k_2^{'} < k_1 \tag{4}$$

In this case, CSL mode would be suppressed through reducing k in the outer slot. In other words, increasing v in the outer slot can also suppress the CSL mode radiation.

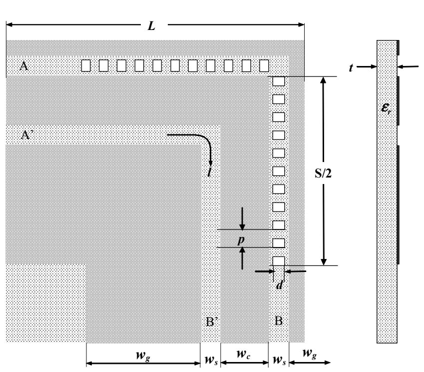
According to the above theory, three techniques can be found out to transform the propagation velocity of signal in the CPW slot. The corrugation techniques used to one or two sides of the slot will form a slow wave structure, and the wave propagating in the slot will be slowed down. The substrate of the CPW is punched with holes at the slot to form a fast wave structure, and the wave velocity in the slot will be speeded up. These corrugation and punching techniques should be used in the inner and outer slots of CPW bend respectively, as mentioned in our previous work [7] and [8]. Besides, the capacitance loading technique also obtains the slow wave effect as the corrugation technique through pasting periodically metallic strip under the background of the slot, and it should also be used in the inner slot to slow down the signal velocity to "wait" for the signal propagating in the outer slot. The velocity compensated CPW bend with the above three techniques are shown as Fig. 1(b)-(d). The steps of producing the velocity compensated CPW bend using the proposed method are concluded as follow: firstly, choose a compensation structure from Fig.1(b)-(d); secondly, determine the design parameters of compensation structure; thirdly, calculate the wave number k_2 or k_2' of this structure by any commercial EM software; fourthly, calculate the length S of the compensation structure according to equations (3) or (4); finally, fabricate the velocity compensated CPW bend according to the compensation structure parameters.



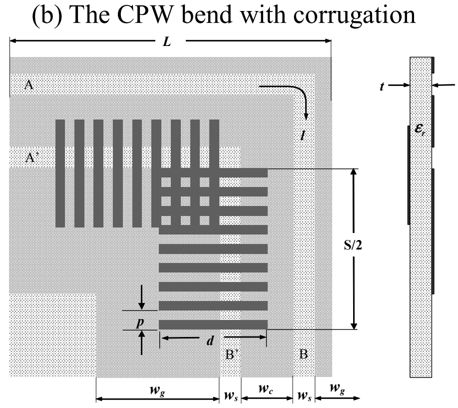
(a) The conventional CPW bend



(b) The CPW bend with corrugation



(c) The CPW bend with hole array

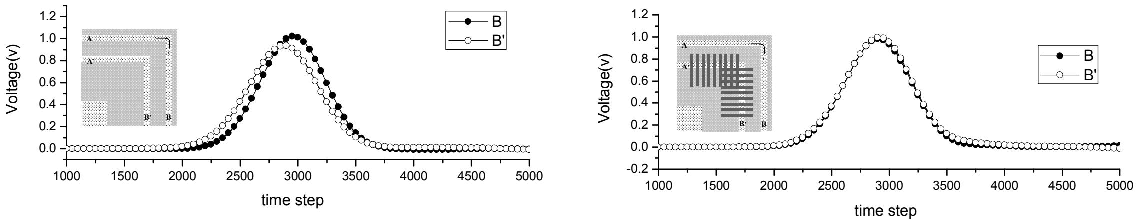


(d) The CPW bend loaded by shunt capacitance

Figure 1 Configuration of the CPW bends with and without compensation

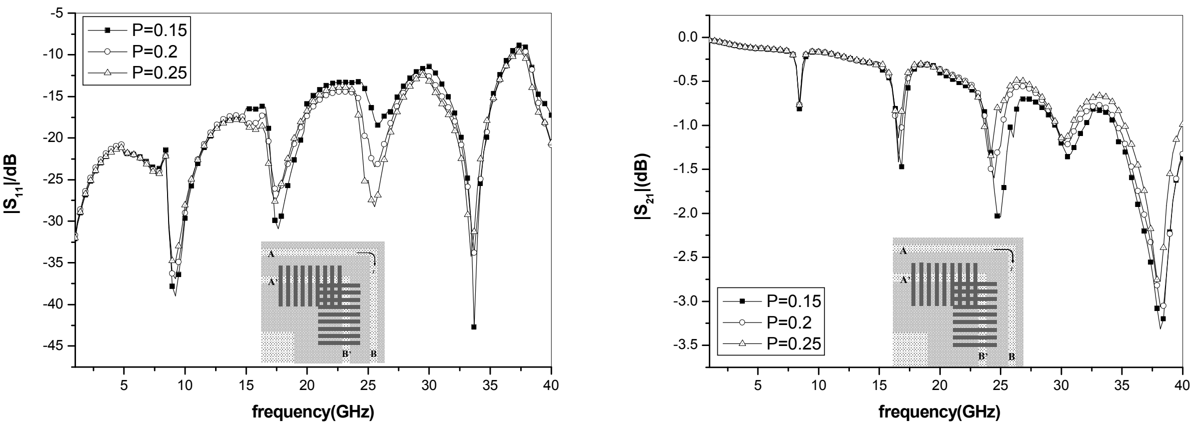
3. Simulation Results

Using the production steps mentioned in last section, the three kinds of velocity compensated CPW bend structure can be confirmed. The simulation and experiment results of the CPW bend with the corrugation and punching techniques have been given in reference [7] and [8], and these results have shown the improvement of the transmission property of CPW bend. In order to further demonstrate the effectiveness of the proposed velocity compensated method, the CPW bend with capacitance loading technique is took as example in this paper. The design parameters of this velocity compensated CPW bend structure are given as follow: the structural parameters of CPW bend without compensation are L=5.225mm, w_c =0.25mm, w_g=1mm, w_s =0.1mm, t =0.625mm, $\varepsilon_r = 9.2$, Z_c=50Ω, the design parameters of periodical metallic strip used in loading are S=5mm, d=0.7mm, p=0.2mm. FDTD method is used to calculate the pulse waveforms propagating in the slots of the CPW bends. The exciting source is the Gaussian pulse with expression: $V(t) = V_0 \exp[-g^2(t-t_{\max})^2]$. In our calculation, V_0=1V, and the Gaussian paramete is selected as g=60×10^9/s. Fig.2 (a) and (b) gives the output pulses waveforms at points B and B' for cases of the CPW bend without and with compensation respectively. This figure shows that for the conventional CPW bend, the two pulses respectively in the inner and outer slot will not arrive at points B and B' at the same time, which is the reason causing radiation of the CSL mode. But after using capacitance loading technique, the two pulses will arrive at B and B' simultaneously, so that CSL mode can be suppressed.



(a) The conventional CPW bend (b) The CPW bend loaded by shunt capacitance

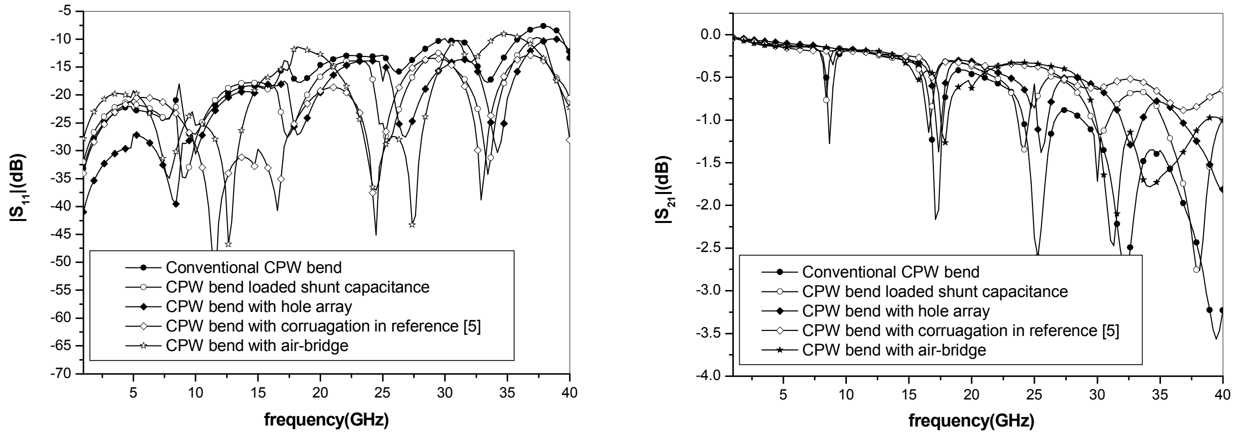
Figure 2 Pulse waveforms in two slots at the output port of CPW bend with and without compensation



(a) Magnitude of S_{11} (b) Magnitude of S_{21}

Figure 3 The effect of the period of shunt capacitance on S parameters of the CPW bend loaded by shunt capacitance

Furthermore, because metallic strip used to compensate the conventional CPW bend are periodic, it's predicted that the period of these structure may play a vital role in the performance of this compensated CPW bend. For this reason, with different periods of the metallic strip, $|S_{11}|$ and $|S_{21}|$ of this compensated CPW bend are figured out by using commercial software HFSS, as shown in Fig.3. Numerical results in Fig.3 show that the transmission performance of the CPW bend loaded by shunt capacitance is improved as the period of the metallic strip increases, but the period of the metallic strip can not increase boundlessly due to the dimension of the CPW bend used in practice. For the case of the CPW bend dimension in this paper, the best period of compensation structure is p=0.25. To verify the effectiveness of the three compensation techniques, the results of the CPW bend with corrugation [7], hole array [8] and air-bridge [2] are also given for comparison, as shown in Fig.4. From Fig. 4, we can draw the following conclusions: first, the average values of reflection of both the CPW bends with the compensation techniques and air-bridge are smaller than those of conventional bend; second, the transmission property of the CPW bend is also significantly improved by the compensation techniques, especially in high frequency band; third, in terms of the compensation effect, air-bridged CPW bend performs better in certain frequency band but worse in the rest band, however, those three kinds of velocity compensated CPW bends perform better in whole investigated frequency band in this paper.



(a) Magnitude of S_{11} (b) Magnitude of S_{21}

Figure 4 *S* parameters of the CPW bends with and without compensation

4. Conclusion

To suppress the CSL mode in asymmetric coplanar waveguide, the method of importing compensation techniques into the conventional CPW is presented, and three kinds of compensation techniques are presented. From above results and discussion, we can observe that these methods can restrain the CSL mode to reduce the radiation and reflection due to the CSL mode. As the conformal property and low cost of the compensation structure, and the suitable frequency band of this structure is broader than that of air-bridged CPW bend, these cheaper velocity compensated CPW bends can replace the relatively costly air-bridged CPW bend.

5. Acknowledgments

This work was supported by the NSFC Projects under Grant No.61201082 and 61102011.

6. References

1. Ma Kuang-Ping, Qian Yongxi, and Itoh Tatsuo, “Analysis and applications of a new CPW-slotline transition,” IEEE Trans. Microwave Theory Tech., vol. 47, NO.4, pp. 426-432, 1999

2. Chungyi Lee, Yaozhong Liu,and Tatsuo Itoh, “The effects of the coupled slotline mode and air-bridge on CPW and NLC waveguide discontinuities,” IEEE Trans. Microwave Theory Tech, vol. 43, NO.12, pp.2759-2764, Dec, 1995

3. Chungyi Lee, Tatsuo Itoh, “The effect of air-bridges on the mode suppression of asymmetrical CPW-fed slot antennas,” Antennas and Propagation Society International Symposium, Vol. 1, pp.338 – 341, June, 1995

4. A. A. Omar, Y L. Chow, and M. G Stubbs, “Effects of air-bridges and mitering on coplanar waveguide 90 degrees bends: theory and experiment,” IEEE MTT-S International Microwave Symposium, vol. 2, pp.823-826, 1993

5. A. A. Omar, Y L. Chow, “Coplanar waveguide with top and bottom shields in place of air-bridges,” IEEE Trans. on Microwave Theory And Techniques, vol. 41, NO. 9, pp. 1559-1563, 1993

6. A. A. Omar, Y L. Chow, “Coplanar waveguide with top and bottom shields in place of air-bridges,” IEEE Trans. on Microwave Theory And Techniques, vol. 41, NO. 9, pp.1559-1563, 1993

7. J. H. Wang, and H. Zhang, “Velocity Compensated coplanar wave guide Bend for Odd-Mode Suppression,” Microwave and Optical Technology Letters, vol. 50, NO.5, pp. 1201-1204, 2008

8. H. Zhang, and J. H. Wang, “Velocity compensated CPW bend for odd mode suppression by hole array,” Chinese Journal of Radio Science, vol. 24, NO. 2, pp. 228-232, 2009